

Challenge (Habanero)

- Q1.** A gamer's high score is 2^5 points. If they double it, what is the new score in terms of exponents?
(A) 2^6
(B) 2^7
(C) 2^8
(D) 2^9
(E) 2^{10}
- Q2.** A coding app has 3^4 levels. If each level has 3^2 puzzles, how many puzzles are there in total, in terms of exponents?
(A) 3^5
(B) 3^6
(C) 3^7
(D) 3^8
(E) 3^9
- Q3.** A data plan costs $15 \cdot 2^x$ dollars per month. If $x = 3$, what is the monthly cost?
(A) 15
(B) 30
(C) 60
(D) 120
(E) 240
- Q4.** What is the simplified form of $(2^2)^3$?
(A) 2^5
(B) 2^6
(C) 2^7
(D) 2^8
(E) 2^9
- Q5.** A digital scoreboard displays a team's score as 8^2 . What is the numerical value of the team's score?
(A) 16
(B) 32
(C) 64
(D) 128
(E) 256
- Q6.** If $x^3 = 64$, what is the value of x ?
(A) 2
(B) 3
(C) 4
(D) 5
(E) 6
- Q7.** A software developer uses 4^x lines of code. If $x = 2$, how many lines of code does the developer use?
(A) 8
(B) 10
(C) 12
(D) 14
(E) 16
- Q8.** What is the value of $(4^2)^{1/2}$?
(A) 2
(B) 4
(C) 8
(D) 16
(E) 32

Open Ended Questions (FRQ)

- Q9.** A gaming company has 3^x users. If they expect 25% more users next quarter, and the number of new users is 3^2 , what is the value of x ? Show all work and calculations.
- Q10.** A programmer uses the rule of exponents to simplify the expression $\frac{(2^3)^2}{(2^2)^3}$. Simplify the expression and show all steps. What is the final numerical value?

Chef's Tips

- Read each question carefully before selecting an answer.
- Use the rules of exponents to simplify expressions.
- Double-check calculations for accuracy.